# Test Prep & Scope Checklist

Full breakdown of the assessment structure and the exact practical scope given verbally in class.

## Assessment Structure (Total: 3 hours, but only ~2hrs needed)

- ■ Web test - True/False - 20 minutes
- ■ Section A - Theory questions (from slides posted on Brightspace) - 40 minutes - covers: explain, give an example, and other question types on data science concepts
- ■ = 1 hour total for Web Test + Theory
- ■ Semester Test 1 - Practical (Jupyter Notebook) - 2 hours

## Platform & Setup

- ■ Go to the lab, open Anaconda
- ■ Anaconda has many packages (Spyder, JupyterLab, PyScript, etc.) - USE: Jupyter Notebook (not JupyterLab, not Colab - no internet in the lab)
- ■ Click 'Launch' on Jupyter Notebook - opens in browser at localhost (similar to Google Colab, ~10-15% different)
- ■ Keep a logbook while working

## Part 1: Dataset Exploration (10 marks)

- ■ Import the libraries you will use
- ■ Load the dataset (dataset will be provided)
- ■ Display the FIRST 5 rows
- ■ Display the LAST 5 rows
- ■ Display the dataset dimensions (shape)
- ■ Display the data types of each column
- ■ Do descriptive statistics (e.g. .describe())

## Part 2: Exploratory Data Analysis (EDA)

- ■ Show summary statistics of the data
- ■ Show the class distribution
- ■ Visualise the data using:
  - ■ Histograms
  - ■ Box plots
  - ■ Correlation map (heatmap)
  - ■ Bar chart

Purpose: observe, understand, and visualise the data.

## Part 3: Missing Values

- ■ Write a script to identify missing values
- ■ Visualise the missing values appropriately
- ■ Decide on a strategy/method to handle the missing values (e.g. imputation) and apply it

## Part 4: Outlier Detection

- Detect outliers in the dataset
- Method 1: Box plot
- Method 2: Interquartile Range (IQR) method
- Identify the outliers clearly

## Part 5: Anomaly Detection

- Identify anomalies in the dataset
- Visualise the detected anomalies

## Part 6: Feature Selection (Data Pre-processing)

- Show which attributes are numerical and which are categorical
- Convert categorical attributes to numerical (encoding technique)
- Standardise the dataset

## Part 7: Train/Test Split

- Split the data into training and testing sets
- NOTE: the exact split ratio (e.g. 80-20, 70-30, 90-10) will be SPECIFIED in the question - read carefully

## Part 8: Logistic Regression Model

- Fit the training data into the Logistic Regression classifier
- Predict using the test set
- Display the prediction probabilities
- Calculate the model accuracy

## Part 9: Model Evaluation (Performance Metrics)

- Classification report
- Confusion matrix
- ROC curve
- ROC-AUC score

## Part 10: Model Interpretation (Explain Your Results)

- Explain the confusion matrix
- Explain the ROC curve
- Explain what the accuracy means for these specific predictions

## Bonus Question (Optional - No Penalty for Skipping, Extra Marks if Done)

- Implement Decision Tree
- Implement Random Forest
- Compare all THREE models: Logistic Regression vs. Decision Tree vs. Random Forest